

Review Feedback — Research paper rough draft.

Date: 2026-07-28

I. Introduction

Grammar / clarity to fix line-by-line

- **1.1:** "*These rapid impressions of which are involuntarily can directly cause real consequential decision-making*" — ungrammatical. Suggest: "*These impressions are involuntary and can shape consequential decisions — who is hired, who gets the loan, who is believed.*"
- **1.1:** "*all derive features from speech in which a growing number of decision-support tools attach consequences to them*" — the "in which... to them" is broken. Rewrite as in the markdown: "*...all derive features from the same acoustic signal that humans interpret socially.*"
- **1.1:** "*social weights of that judgment*" — should be "*social weight*" (singular).
- **1.3:** "*Frame stability is thereby a required property for a deployer or regulatory*" — "regulatory" is an adjective. Should be "*deployer or regulator*".
- **1.4:** The paragraph starting "*The Random Forest and Logistic Regression baselines...*" is a sentence fragment (no main verb). Add "*We compare five classifiers: ...*".
- **1.4:** "*We address both problems of the Trustworthy intent in Speech Corpus*" — should be "*We address both problems **using** the ...*".

Content losses vs. the markdown

- The markdown's contributions bullet list gives concrete numbers (73.4/71.9%, 76.5±2.5%, 5.0pp gap, 0.89pp std). The PDF's 1.6 collapses these into a running paragraph, losing the punch. Restore the bulleted format with numbers — reviewers scan contribution lists first.
- The markdown opens with a stronger, tighter first sentence: "*The human voice is among the richest social signals we produce.*" The PDF's "*Much of human life revolves around communication...*" is a generic hook. Use the markdown version.
- The markdown Introduction cites Ganin et al. (2016) properly for the DANN paradigm and includes the shortcut-amplification argument. The PDF is weaker on why non-linear models introduce a new bias risk — this motivation belongs before you introduce DANN.

Structural

- 1.6 has "*Reproduction of the published baselines & A controlled comparison of five classifiers*" squashed into a single bullet. Split them.
 - 1.7 Roadmap uses Roman numerals but the rest of the paper uses Arabic — fix once you pick a scheme.
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III. Dataset (labelled "4." in the draft)

Structural

- The heading is "*4. Dataset and Features.*" but the roadmap says Section III. Fix to Section III.

- **Table 1 has spilled across a page break** (page 10 → 11) and lost its formatting — the second half becomes free-flowing text: "*Black Younger 11 9 20 Black Older 5 3 8 South Asian Younger 10 10 20...*". Wrap Table 1 in a `\begin{table}` (LaTeX) or "Keep with next" (Word) so it never breaks mid-body.

Content losses vs. the markdown

- The PDF describes the acoustic feature set as "*fundamental frequency (mean and standard deviation of F_0), voice-quality measures ... sentence duration, and long-term average spectrum descriptors (LTAS mean, standard deviation and slope)*". This lists only ~10 features but then says all 60 columns are used. The markdown's **five families** breakdown (Pitch 10, Voice-quality 12, Formant 14, Spectral 8, LTAS 6) is much more informative and accounts for all 60. **Port the five-family paragraph in verbatim.**
- The markdown flags the older-cell imbalance visually via **Fig. 2** and describes stratification as the mitigation. The PDF mentions the imbalance but does not reference the figure.
- The PDF says older-adult recruitment used "*posters and word of mouth*"; the markdown says "*in-person word-of-mouth*". Verify which matches the source paper and unify.

Prose issues

- **4.1.1:** "*All experiments conducted are attributed to the Trustworthy Intent in Speech (TIS) Corpus*" — "attributed to" is wrong here. Use the markdown's "*All experiments are conducted on ...*".
- **4.1.3:** The last sentence spills from page 12 to page 13 as "*Stratifying on intent alone, as the source paper effectively does under leave-one-speaker-out cross-validation, lets the test set's ethnicity composition drift...*" — long and hard to parse. Break into two sentences per the markdown.
- **4.2** "*Original analysis recap*" — good addition not in the markdown, but the LOSO / accuracy numbers here overlap with material also in Section V. Decide where it lives and remove the duplicate.
- **4.3** "*A note on label validity*" — this same caveat appears again in Discussion (VI-C limitation #2). Either keep it here as motivation and drop it from Discussion, or keep it only in Discussion. The current draft says it twice.

Missing elements from the markdown

- No mention of `Speech_dataset_characteristics.csv` as the exact input file (the markdown names it).
- No mention of the LOSO caveat "*typically reduces all reported accuracies by approximately 1 to 3 percentage points.*" — this is useful for reader calibration.

VI. Discussion (labelled "9." in the draft)

Blocking issues

- **Section 9.3 is truncated** — ends at "*Seeds:*" mid-sentence. Missing: the full 6-row Limitations table (Table 10 in the markdown), the three limitations discussed in prose (small N, self-rated labels, DANN leakage), and the closing paragraph about the sixth methodological limitation being the interesting research direction.
- **Missing "What Worked" subsection structure.** The markdown organises VI-A around four distinct findings (accuracy lift, fairness lift, reproducibility from adversarial training, acoustic findings replicate). The PDF's 9.1 collapses them into one long paragraph — reader loses the four-part scaffold. Restore bold sub-labels.

Content losses vs. markdown

- The markdown VI-A explicitly notes "*Acoustic findings replicate*" (F0 mean, F0 SD, HNR, shimmer, CPP as dominant cues per Random Forest Gini importance). This independent replication of the source paper's acoustic finding is a *fourth* contribution to the Discussion and is missing from the PDF entirely.
- **Fig. 13 discussion is diluted.** The markdown VI-B gives the (accuracy, gap) plane a clear geometry: "lower-right is good", "green arrow points there", RF/LR in upper-left, ANN upper-right, CNN/DANN lower-right. The PDF has the numerical facts but not the visual story. Restore the geometry-first framing.
- The markdown notes that Fig. 13 "*marks the source paper's Random Forest LOSO result as a black star at (71%, 5.0pp)*" — key visual anchor for the headline claim. Missing from the PDF.

Prose issues

- **9.1** opens "*All three neural models have outperformed both baselines*" — tense drift; use simple past: "*outperformed*" only. And "*our largest single beneficiary*" is awkward; markdown uses the cleaner "*The biggest beneficiary was the South Asian group*".
- **9.1:** "*DANN-Trust posits the most consistent per-group profile*" — "posits" is the wrong verb. Use "*delivers*" or "*achieves*".
- **9.2:** The distinction between **behavioural fairness (accuracy parity)** and **representational fairness (invariance)** is well made — keep this exactly as in the markdown (it is the strongest paragraph in the section). Cite Hardt et al. (2016) — the PDF does so; make sure the reference is in the bibliography.
- **9.2** spills onto page 35 with "*groups roughly equally in accuracy on this corpus. We cannot claim however that it no longer represents ethnicity internally.*" — good sentence, but hyphenate "however" correctly: "*We cannot claim, however, that ...*".

Structural

- 9.3 will need Table 10 (the six-row limitations table from the markdown). Confirm the mitigations column matches what you actually plan to do — items 4 and 5 (λ sweep, k-fold CV) are the two cheapest to actually run before submission and would strengthen the paper if you did them, rather than list them as future work.

Priority order for revision

1. Fix PDF export (word-per-line spacing) and delete the "Plan" / meeting-note front matter.
 2. Complete the truncated 9.3 Limitations section (port Table 10 from the markdown).
 3. Fix Table 1 formatting so it does not spill across the page break.
 4. Reconcile section numbering (Roman vs. Arabic) across roadmap, headings, and cross-references.
 5. Restore the four-part "What Worked" scaffold in the Discussion and the five-family feature breakdown in the Dataset.
 6. Line-level grammar pass on the Introduction — several sentences do not parse.
 7. Unify citation style (numeric IEEE-style is the target given the existing [1], [14], [17] in Dataset).
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The markdown files under **MySolution/Paper/** are close to ready. The main task is bringing the PDF back in line with them, not rewriting from scratch.